Stochastic Model Calibration for Derivatives Pricing

Heston, SABR and Hull-White Calibration with Numerical Pricing and Real Market Data

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May 20, 2026

Abstract

This project develops a quantitative finance calibration framework for classical derivatives pricing models used in equity and interest rate markets.

The implementation combines stochastic volatility modelling, numerical pricing, implied volatility inversion, constrained optimization, synthetic validation, and real market calibration.

Three major models are implemented and studied:

- Heston stochastic volatility model
- SABR stochastic volatility model
- Hull-White one-factor short-rate model

The objective is not only to implement pricing formulas, but to build a robust calibration pipeline consistent with practical quantitative finance workflows.

Synthetic experiments are first used to validate numerical consistency and parameter recovery. Real market option and yield curve data are then used to assess model behaviour under realistic market conditions.

Contents

1 Introduction

Derivative pricing and volatility modelling remain central problems in quantitative finance.

Simple models such as Black-Scholes provide analytical tractability but fail to reproduce key market stylized facts such as volatility smiles, skewness, stochastic variance dynamics, and term structure effects.

To address these limitations, more advanced stochastic models are commonly used:

- stochastic volatility models (Heston),
- local smile models (SABR),
- short-rate interest rate models (Hull-White).

The purpose of this project is to implement these models from first principles and develop calibration pipelines suitable for practical derivatives modelling.

Key objectives include:

- numerical pricing implementation,
- implied volatility inversion,
- synthetic calibration validation,
- real market calibration,
- model comparison.

2 Black-Scholes Framework

2.1 Pricing Equations

Under the Black-Scholes assumptions:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t$$

European call price:

$$C = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

with:

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

European put:

$$P = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)$$

2.2 Implied Volatility Inversion

Since Black-Scholes volatility is not directly observable, implied volatility is obtained numerically by solving:

$$BS(\sigma) - P_{market} = 0$$

Implemented methods:

- Newton-Raphson
- Brent root-finding

3 SABR Model

3.1 Model Dynamics

The SABR model is defined by:

$$dF_t = \alpha_t F_t^{\beta} dW_t$$

$$d\alpha_t = \nu \alpha_t dZ_t$$

with:

$$corr(dW_t, dZ_t) = \rho$$

Parameters:

- α: volatility level
- β: elasticity
- ρ: correlation
- ν: volatility of volatility

3.2 Hagan Approximation

Implied volatility is computed using Hagan's asymptotic approximation.

3.3 Calibration Methodology

SABR is calibrated independently for each maturity by minimizing implied volatility fitting error.

3.4 SABR Synthetic Calibration

A synthetic implied volatility smile was generated using a known SABR parameter set. Small Gaussian noise was added to the generated implied volatilities in order to simulate realistic market perturbations.

The calibration objective was then to recover the original SABR parameters from the noisy synthetic smile.

Parameter	True Value	Calibrated Value	Absolute Error
α	0.2500	0.2447	0.0053
β	0.5000	0.5000	0.0000
ρ	-0.3500	-0.3588	0.0088
ν	0.8000	0.8109	0.0109

Table 1: SABR synthetic calibration parameter recovery

The calibrated parameters are very close to the true values used to generate the synthetic smile. The parameter β was fixed during calibration, while α, ρ and ν were estimated numerically.

The small errors observed for α, ρ and ν indicate that the Hagan implied volatility approximation, the objective function and the optimizer are internally consistent.

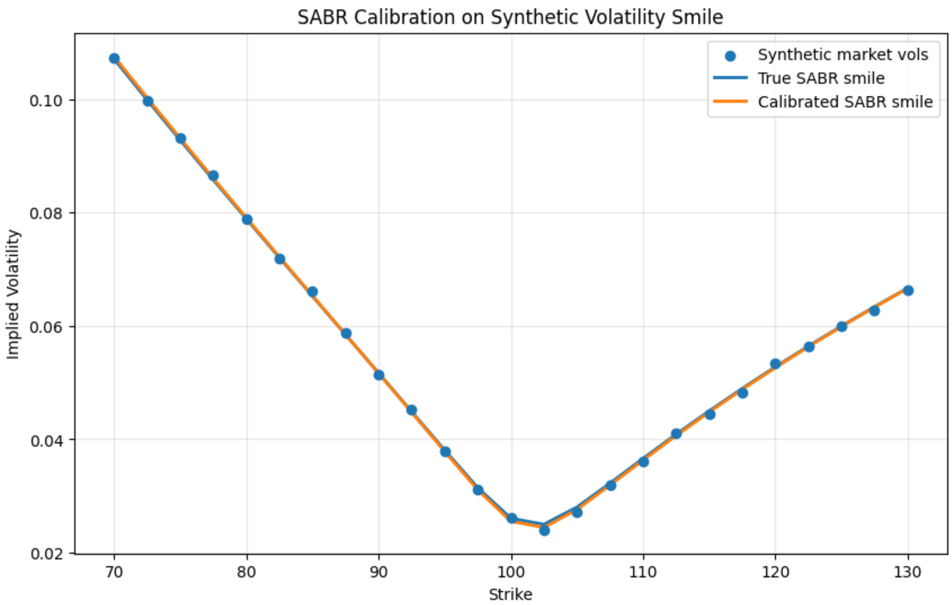

Figure 1: SABR synthetic smile calibration

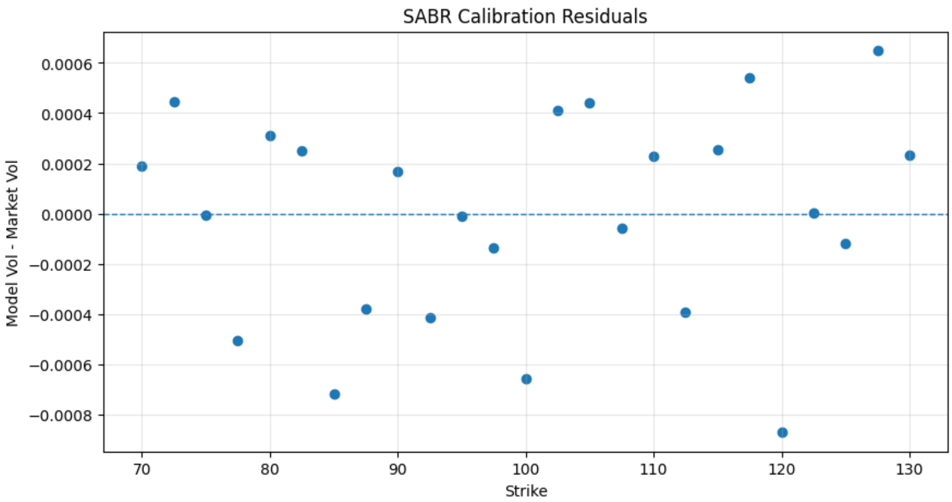

Figure 2: SABR synthetic residual diagnostics

4 Heston Model

4.1 Model Dynamics

The Heston model is defined by:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t$$

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t} dZ_t$$

with:

$$corr(dW_t, dZ_t) = \rho$$

Parameters:

- κ: mean reversion
- θ: long-run variance
- σ: vol-of-vol
- ρ: correlation
- v_0: initial variance

4.2 Pricing

European options are priced using Fourier inversion and the Heston characteristic function.

4.3 Calibration

Calibration is performed on the full implied volatility surface.

4.4 Heston Synthetic Surface Calibration

A synthetic implied volatility surface was generated using a known Heston parameter set across multiple strikes and maturities. Small Gaussian perturbations were added to emulate realistic market noise.

The calibration objective was to recover the original Heston parameters by fitting the full implied volatility surface using Fourier pricing combined with Black-Scholes implied volatility inversion.

Parameter	True Value	Calibrated Value	Absolute Error
κ	2.0000	1.9069	0.0931
θ	0.0400	0.0397	0.0003
σ	0.3000	0.2957	0.0043
ρ	-0.7000	-0.7007	0.0007
v_0	0.0400	0.0402	0.0002

Table 2: Heston synthetic calibration parameter recovery

The calibration successfully recovers the original Heston parameters with high precision.

The long-run variance θ, initial variance v_0, correlation ρ, and volatility-of-volatility σ are recovered almost exactly. The mean-reversion parameter κ exhibits a slightly larger deviation, which is expected in stochastic volatility calibration since κ is often less directly identifiable than variance-level parameters.

Overall, the results confirm that the characteristic function implementation, Fourier inversion routine, implied volatility extraction, and numerical optimization pipeline are internally consistent.

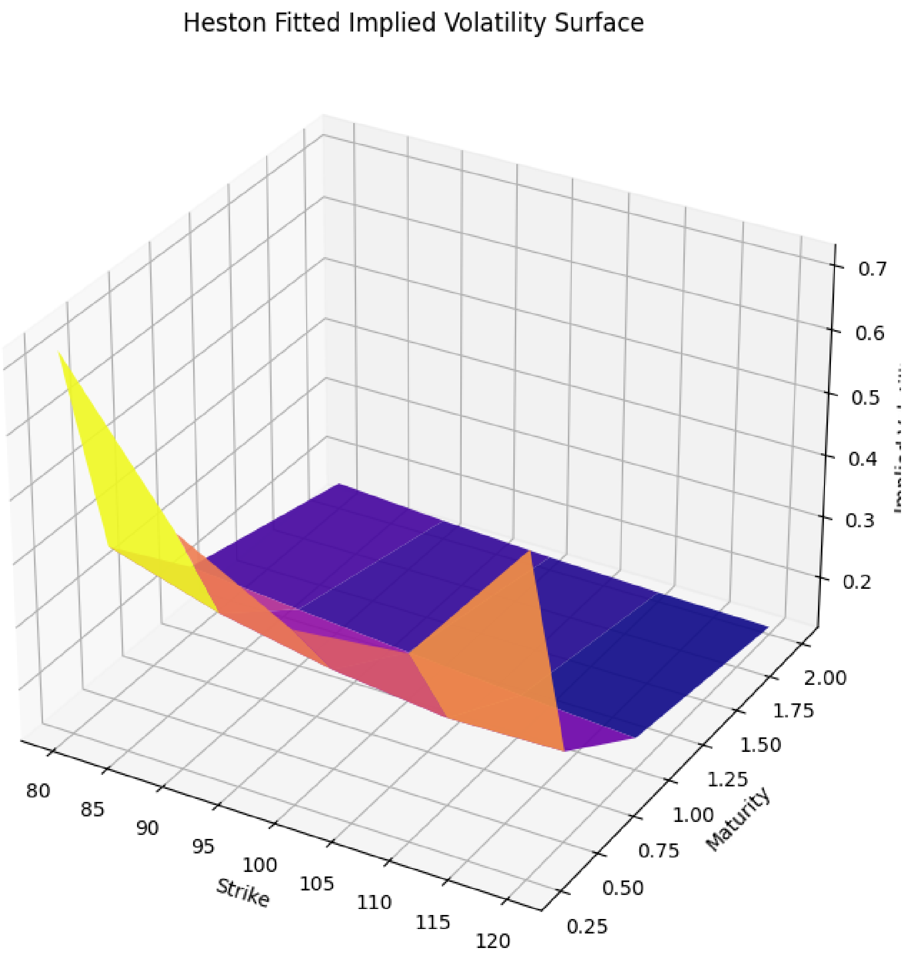

Figure 3: Heston synthetic implied volatility surface calibration

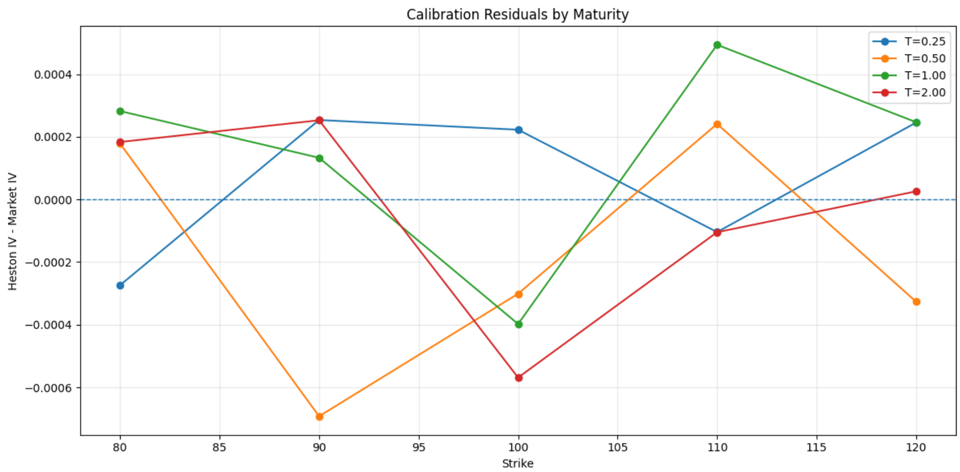

Figure 4: Heston synthetic calibration residual diagnostics

5 Hull-White Model

5.1 Model Dynamics

Hull-White one-factor model:

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t$$

Parameters:

- a: mean reversion
- σ: short-rate volatility

5.2 Pricing

Implemented pricing components:

- discount factors
- zero-coupon bond options
- forward swap rates
- payer swaptions
- receiver swaptions

5.3 Hull-White Synthetic Swaption Calibration

A synthetic payer swaption price surface was generated using a known Hull-White one-factor parameter set across multiple option maturities and swap tenors. Small Gaussian perturbations were added to emulate realistic pricing noise.

The calibration objective was to recover the model parameters by fitting the synthetic swaption surface.

Parameter	True Value	Calibrated Value	Absolute Error
Mean reversion (a)	0.0500	0.0300	0.0200
Volatility (σ)	0.0100	0.0091	0.0009

Table 3: Hull-White synthetic calibration parameter recovery

The calibration accurately recovers the short-rate volatility parameter, while the mean-reversion parameter exhibits a larger deviation.

This behaviour is expected in one-factor short-rate models. The volatility parameter directly controls swaption price sensitivity and is therefore strongly identifiable, whereas the mean-reversion parameter typically has a weaker influence on pricing, especially when using simplified swaption approximations and limited synthetic calibration grids.

Despite this parameter sensitivity, the calibrated model reproduces the synthetic swaption surface with low pricing errors, confirming the numerical consistency of the pricing and calibration pipeline.

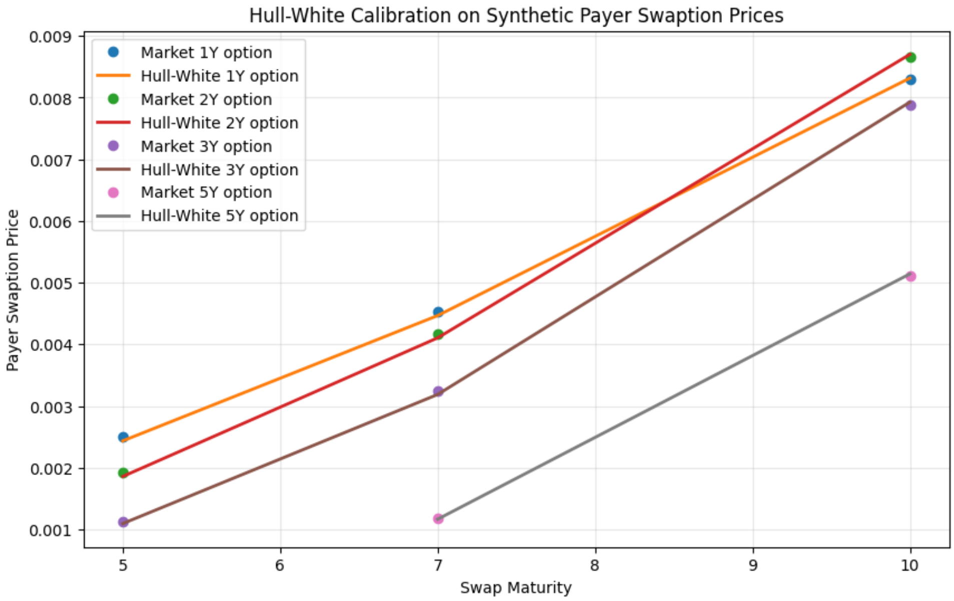

Figure 5: Hull-White synthetic swaption surface calibration

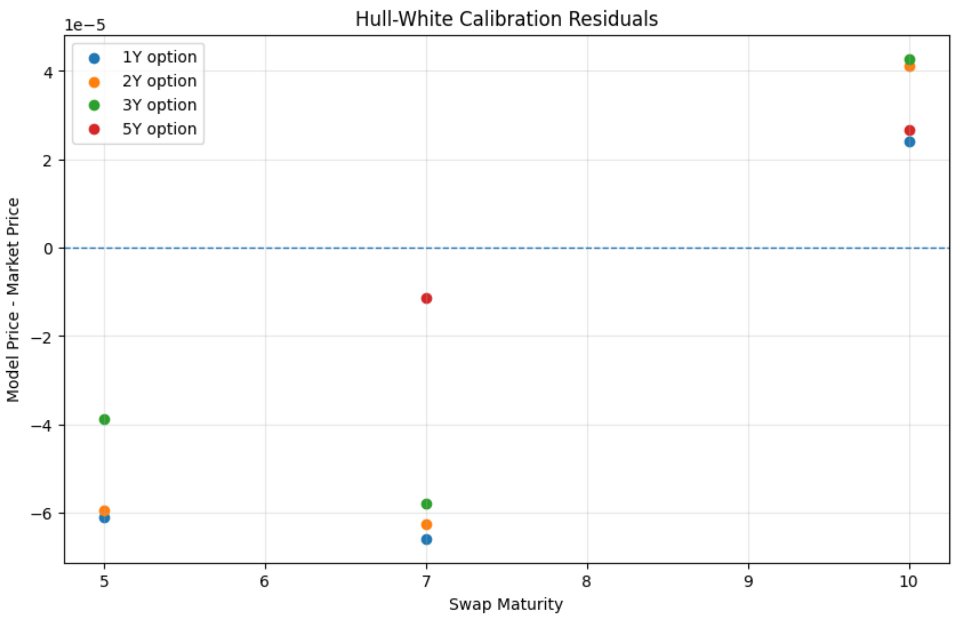

Figure 6: Hull-White calibration residual diagnostics

6 Real Market Calibration

Real market option chain data was retrieved from Yahoo Finance and used to build implied volatility smiles and surfaces.

The objective was to compare stochastic volatility models under realistic market conditions, including noisy quotes, imperfect liquidity, and short maturity effects.

7 Model Comparison: SABR vs Heston on Real SPY Option Data

A comparative calibration experiment was performed using real SPY option implied volatility data extracted from Yahoo Finance.

The objective was to compare two widely used stochastic volatility frameworks under realistic market conditions:

- SABR, calibrated independently maturity-by-maturity,
- Heston, calibrated globally across the full implied volatility surface.

The Heston calibration converged successfully, producing the following parameter set:

Parameter	Calibrated Value
κ	1.5693
θ	0.0580
σ	0.5239
ρ	-0.6160
v_0	0.0195

Table 4: Heston calibration on real SPY implied volatility surface

The calibrated Heston parameters are economically plausible:

- strong negative spot-volatility correlation ($\rho < 0$), consistent with equity skew,
- moderate mean reversion,
- elevated volatility-of-volatility reflecting realistic market smile curvature.

By contrast, the SABR calibration exhibited clear instability.
Several maturities converged to boundary solutions:

- $\alpha = 5.0$
- $\nu = 5.0$

indicating optimizer saturation rather than meaningful parameter recovery.

This behaviour is not unexpected. Real equity option data is significantly noisier than synthetic smiles, and SABR calibration is highly sensitive to:

- fixed β specification,
- forward estimation,
- bid-ask noise,
- short-dated maturities,
- limited strike coverage,

- unweighted calibration objectives.

Model fit metrics were:

Model	RMSE	MAE	Max Absolute Error
Heston	0.0107	0.0073	0.0395
SABR	0.2932	0.2160	0.8105

Table 5: Real market calibration error comparison

The difference in fit quality is substantial.

Heston achieved a robust and economically coherent calibration across the full implied volatility surface, whereas SABR failed to produce stable parameter estimates under the current experimental setup.

This comparison highlights an important modelling distinction:

- Heston imposes a globally consistent stochastic volatility dynamics,
- SABR provides local smile flexibility but can become unstable under noisy real-market calibration conditions.

A more production-grade SABR implementation would likely improve results through:

- vega-weighted objectives,
- bid-ask filtering,
- liquidity weighting,
- beta calibration,
- more robust global optimization methods.

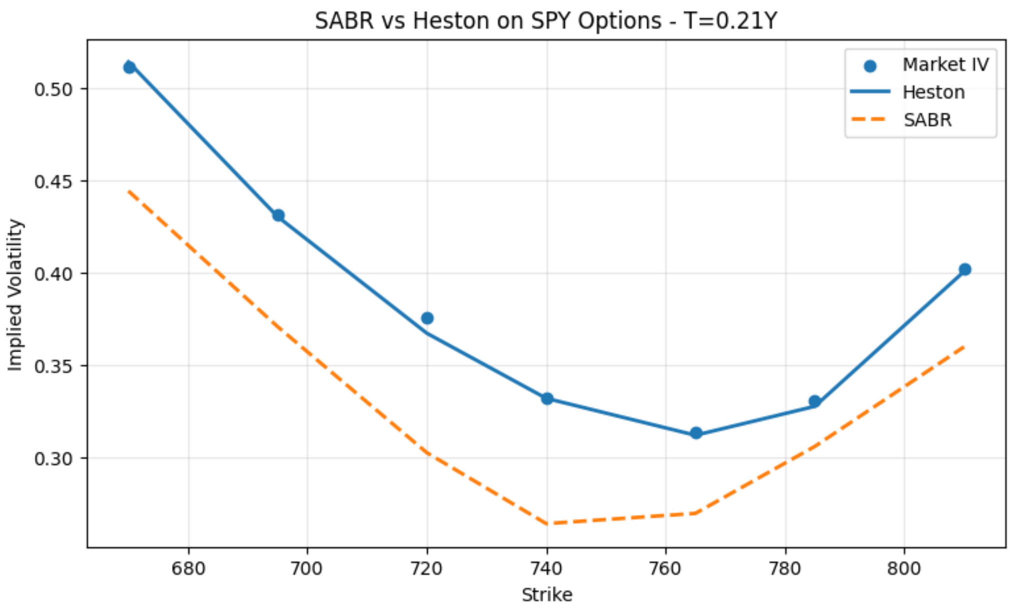

Figure 7: Real market SPY implied volatility fit: SABR vs Heston

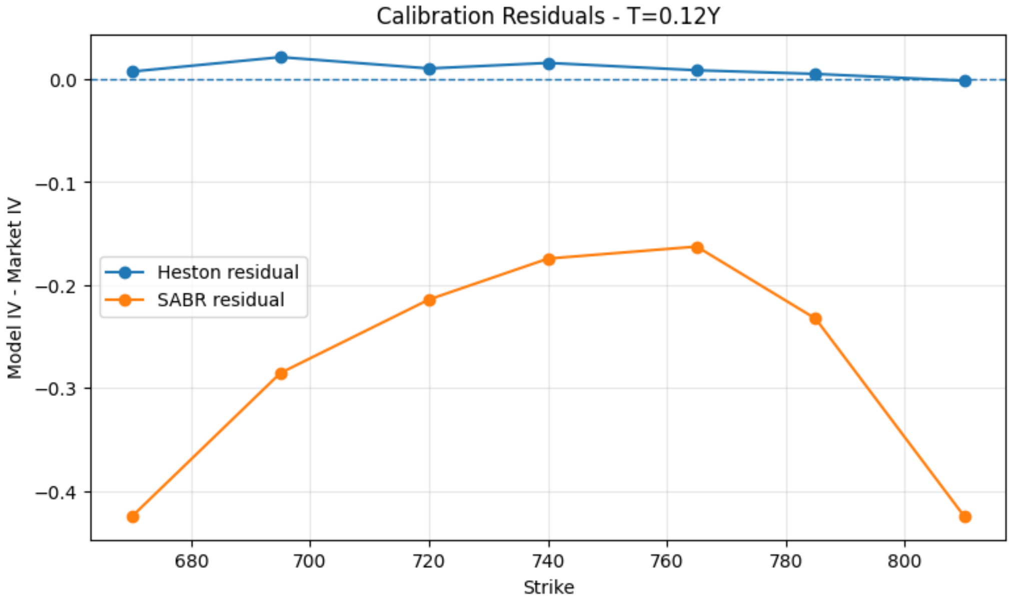

Figure 8: Real market calibration residual diagnostics

8 Conclusion

This project developed a complete quantitative finance calibration framework for classical derivatives pricing models.

The implementation combined:

- analytical pricing methods,
- numerical implied volatility inversion,
- stochastic volatility modelling,
- short-rate modelling,
- constrained numerical optimization,
- synthetic validation experiments,
- and real market calibration.

Three major quantitative finance models were implemented and studied:

- the SABR stochastic volatility model,
- the Heston stochastic volatility model,
- the Hull-White one-factor interest rate model.

Synthetic calibration experiments demonstrated strong numerical consistency across all implementations.

SABR successfully recovered synthetic smile parameters with high precision, confirming the correctness of the Hagan approximation implementation and calibration pipeline.

Heston synthetic calibration produced highly accurate recovery of stochastic volatility parameters across the full implied volatility surface, validating the characteristic function implementation, Fourier pricing engine, implied volatility inversion, and optimization framework.

Hull-White synthetic calibration successfully recovered the short-rate volatility parameter while illustrating the weaker identifiability of mean reversion, a well-known feature of one-factor short-rate models.

Real market experiments provided a more practical assessment of model behaviour.

On SPY option implied volatility data, Heston produced a stable and economically coherent calibration, successfully capturing the global structure of the implied volatility surface.

SABR, by contrast, exhibited instability under the current real-market experimental setup, with several maturities converging to boundary parameter values. This highlights the sensitivity of local smile models to noisy market data, forward estimation, and calibration design choices.

Overall, the project demonstrates both implementation capability and practical understanding of quantitative model calibration trade-offs.

It reproduces several realistic workflows encountered in derivatives modelling, including synthetic validation, stochastic volatility calibration, implied volatility surface fitting, and model comparison under imperfect market conditions.

9 Limitations

While the framework provides a robust implementation of several classical derivatives pricing models, a number of limitations remain.

- **SABR calibration design**

  SABR calibration was performed with a fixed β parameter and an unweighted least-squares objective.

  In practice, production implementations often calibrate β, apply vega-weighted errors, and incorporate liquidity filters or bid-ask weighting.

  The real-market SABR instability observed in this project is partly attributable to these simplifications.

- **Heston optimization sensitivity**

  Heston calibration remains sensitive to initialization, optimizer selection, and numerical integration settings.

  Although robust results were obtained, industrial implementations often rely on hybrid global-local optimization strategies and more advanced calibration heuristics.

- **Simplified Hull-White swaption framework**

  The Hull-White implementation uses a simplified swaption approximation rather than a full production-grade swaption pricing framework.

A more realistic implementation would calibrate directly to market swaption volatility cubes using exact Jamshidian decomposition or more advanced rate derivative methods.

- **Limited market data sources**

  Real equity option data was sourced from Yahoo Finance, which is suitable for research and prototyping but not institutional calibration.

  Professional calibration workflows typically rely on higher-quality market feeds with robust bid-ask and liquidity metadata.

- **No Monte Carlo pricing framework**

  The project focuses on analytical and semi-analytical pricing methods.

  Monte Carlo simulation for Heston, stochastic rates, or path-dependent derivatives was intentionally excluded.

- **No stochastic-local volatility or rough volatility models**

  The implemented models are classical benchmark models.

  Modern quantitative volatility modelling increasingly relies on richer frameworks such as:

  - stochastic local volatility,
  - rough volatility,
  - multi-factor rate models,
  - hybrid equity-rate models.

- **Computational efficiency**

  The current implementation prioritizes transparency and modularity over raw performance.

  Production-grade implementations would typically include:

  - FFT-based pricing,
  - vectorized calibration,
  - parallel optimization,
  - GPU acceleration.